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Postural control in young and elderly adults when stance is perturbed: dynamics

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PMID: 8850638 DOI: 10.1016/0021-9290(95)00052-6

Abstract

Responses in maintaining or restoring standing balance were measured in 24 healthy young and 15 healthy elderly adults (mean ages 26 and 72) under four task conditions: two involving self-generated motions and two involving imposed disturbances. The two primary objectives of the study were to quantify the whole-body dynamics of these responses and to identify any age related differences in those dynamics. Response dynamics were analyzed using a seven-link biomechanical model. In terms of approximate population-mean values, maximum whole-body center of mass (CM) excursions ranged to 3 cm, maximum center of support-surface reaction (CR) excursions ranged to 8 cm, vertical reaction force changes ranged to 50 N, anteroposterior support surface reactions ranged to 30 N, maximum joint torques used per side ranged to 20 Nm and peak angular momenta about a transverse axis through the ankles ranged to 6 kg m2 s-1. The elderly adults, compared to the young tended to exhibit higher-frequency oscillations in excursions and larger horizontal excursions of their CM and CR, tended to develop larger support surface reactions and use larger response joint torques, and tended to arrest less of their angular momentum in their first cycle of response during the two imposed-disturbance tasks. Only some of these tendencies proved statistically significant. The results suggest that healthy elderly subjects with no apparent musculoskeletal or neurological impairments differ from healthy young adult subjects in their responses to modest perturbations of upright stance. However, the differences are generally not large and their magnitudes are perturbation-specific.

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